
BEHAVIORAL SHIFTS IN DIGITAL FINANCE: HOW E-PAYMENT INFLUENCES CONSUMER SPENDING AND FINANCIAL LITERACY

Faizal Rizky Yuttama

Universitas Harapan Bangsa, Indonesia
e-mail: faizal@uhb.ac.id

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ABSTRACT

Objective: Examine how e-payment usage influences consumer spending and cash management, and assess the moderating role of financial literacy in an emerging-market context. **Research Design & Methods:** Quantitative survey of active e-payment users in Central Java, Indonesia. Constructs were measured with Likert-scale items. Relationships among e-payment usage, consumer behavior, cash management, and financial literacy (moderator) were analyzed using PLS-SEM with established validity and reliability procedures. **Findings:** E-payment facilitates more frequent and effortless purchasing by lowering psychological frictions at checkout, which can elevate impulsive spending. At the same time, its built-in records and dashboards support better oversight of income–expense flows, budgeting, and planning. Financial literacy shapes these outcomes: higher literacy helps curb excessive spending and strengthens the use of app features for monitoring, budgeting, and goal setting. Overall, the impact of e-payment is dual and contingent on users’ financial knowledge and self-regulation. **Implications & Recommendations:** Pair cashless initiatives with digital financial-literacy programs. Embed in-app nudges such as spend alerts, weekly summaries, category caps, and goal tracking. Tailor guidance for low-literacy segments and encourage responsible defaults. **Contribution & Value Added:** Provides emerging-market evidence clarifying when and how e-payments promote or undermine financial discipline, integrates behavioral-finance and financial-literacy perspectives via moderation analysis, and offers actionable design and policy insights to enhance consumer financial well-being.

Keywords: behavioral finance; cash management; consumer behavior; e-payment; financial literacy.

JEL codes: G53, G41, E42

Article type: research paper

INTRODUCTION

The rapid evolution of financial technology (FinTech) has significantly transformed global payment systems, shifting consumer transactions from cash-based to digital platforms. Among these advancements, electronic payment (e-payment) systems have emerged as a dominant financial tool, enabling seamless, convenient, and secure transactions. With the growing emphasis on cashless economies, many governments and financial institutions worldwide are actively promoting digital payment adoption to enhance financial inclusion and economic efficiency.

Indonesia has experienced a substantial rise in e-payment adoption, driven by technological advancements, increased smartphone penetration, and government-led digital financial initiatives. According to [Bank Indonesia \(2023\)](#), digital transactions in Indonesia grew by 20.3% in 2022, reaching IDR 395.8 trillion (USD 26.3 billion), with e-wallet transactions accounting for 45% of total retail payments. In Central Java, the adoption of e-wallets and mobile banking has surged, particularly among

young consumers and small businesses, reflecting a broader shift toward cashless financial behavior. While this transition offers numerous advantages, such as enhanced convenience, transaction security, and financial tracking, concerns have emerged regarding its impact on consumer spending habits and financial management practices.

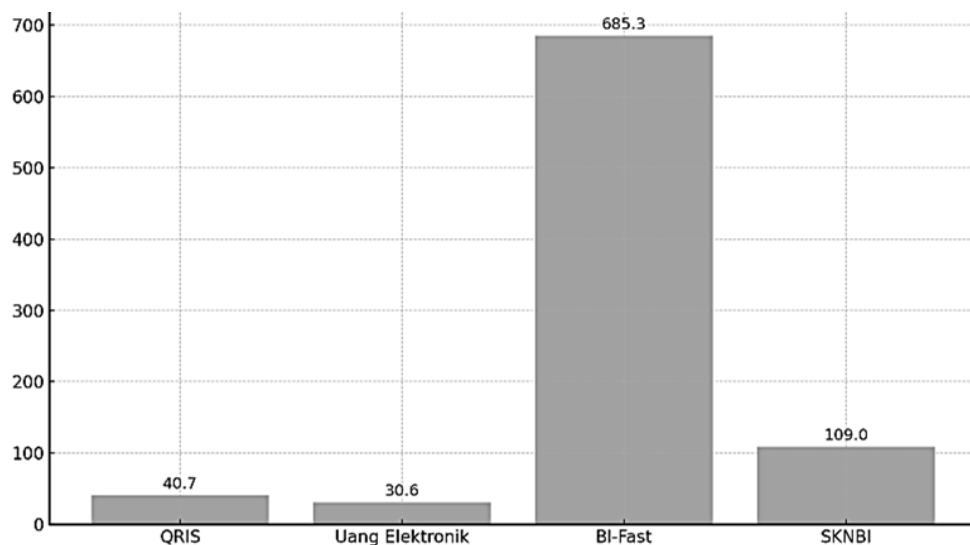


Figure 1. E-Payment Transaction Value Growth Trend in Central Java (2023-2024)

Source: Bank Indonesia, 2024

Figure 1 illustrates a significant surge in the use of digital payment systems during 2023 to 2024. Notably, QRIS (Quick Response Code Indonesian Standard) transactions experienced rapid growth, with the transaction value increasing by 385.12% year on year, reaching a total of IDR 40.7 trillion from more than 411 million transactions. The number of QRIS users also rose sharply to 7.63 million, with over 3.6 million merchants most of whom are from the MSME (Micro, Small, and Medium Enterprises) sector indicating that digitalization has extensively penetrated the micro-business sector. On the other hand, electronic money (e-wallet) transactions also recorded positive growth, with a transaction value of IDR 30.6 trillion, representing an increase of approximately 18.73% compared to the previous year. Other infrastructures, such as BI-Fast and SKNBI, further strengthened the digital payment ecosystem in this province. BI-Fast recorded transaction values reaching IDR 685.3 trillion, while SKNBI transactions amounted to IDR 109 trillion. Equally significant, the city of Semarang reported over 620,000 users of the DANA digital wallet, with a 20% annual growth in transactions. This data reflects a growing tendency among the people of Central Java both individual consumers and business actors to adopt e-payment services in their daily economic activities. This phenomenon indicates not only a shift in payment preferences but also a broader transformation in financial behavior towards digital practices, which must be accompanied by improved digital financial literacy to ensure productive e-payment usage and mitigate long-term financial risks.

The widespread adoption of e-payment systems has raised concerns about their influence on consumer behavior and personal financial management. Digital payments have been linked to increased consumer spending, as they reduce the psychological barriers associated with cash transactions, leading to higher impulsivity and lower spending awareness, as noted by [Hampson et al. \(2021\)](#). On the other hand, e-payment systems offer enhanced financial tracking tools, allowing users to monitor their expenses and manage their cash flow effectively, which aligns with findings from [Liu & Zhang \(2021\)](#). However, the extent to which e-payment encourages excessive spending or improves financial discipline remains inconclusive, particularly in emerging markets like Indonesia.

While prior studies have largely treated e-payment as a direct driver of spending or cash management outcomes, this relationship may not be uniform across all consumers. The influence of e-payment on financial behavior depends substantially on individual differences in financial literacy. Financial literacy acts as a moderating mechanism that determines how consumers interpret, evaluate, and respond to the convenience and automation of digital transactions ([Lusardi & Mitchell, 2014](#); [OECD, 2018](#)).

Individuals with higher literacy levels are more likely to leverage e-payment functions such as transaction records, budget reminders, and spending analytics for informed decision-making and self-control (Fernandes et al., 2014; Zhavoronok et al., 2022). Conversely, those with lower financial literacy may perceive digital payment ease as an invitation to spend impulsively, lacking the awareness to monitor accumulated expenses (Hampson et al., 2021; Ngo et al., 2024). Therefore, financial literacy does not merely accompany e-payment usage but shapes the direction and intensity of its behavioral consequences, making it a critical factor in determining whether e-payment adoption enhances financial discipline or leads to overspending (Graña-Alvarez et al., 2024; Liu & Zhang, 2021).

Moreover, financial literacy has been identified as a key factor influencing financial decision-making, yet its role in moderating the effects of e-payment usage on consumer behavior and cash management remains underexplored. Individuals with higher financial literacy may utilize e-payment tools for better financial control, whereas those with limited financial knowledge may struggle with over-spending and poor money management (Graña-Alvarez et al., 2024). Understanding how financial literacy moderates the relationship between e-payment usage, spending behavior, and cash management is crucial for designing effective financial policies and consumer education programs.

Several studies have examined the impact of digital payments on consumer behavior and financial management, yet key gaps remain in the literature. Usman et al. (2025) explored the relationship between digital payment adoption and consumer spending, highlighting that e-payment facilitates impulse buying but did not assess its impact on financial planning or savings behavior. Laukkanen (2016) investigated consumer adoption of mobile and internet banking but did not examine how digital payment behavior affects personal cash management. Liu & Zhang (2021) discussed the opportunities and challenges of digital payments in financial management, yet their study lacked an empirical analysis of consumer financial habits in developing economies like Indonesia. Furthermore, Hampson et al. (2021); Ly & Ly (2024) provided theoretical and empirical insights into how digital payment mechanisms influence spending behavior and trust, but limited research has explored the moderating role of financial literacy in shaping these effects. This study addresses this gap by examining how financial literacy influences the relationship between e-payment usage, spending behavior, and cash management in Central Java, offering new insights into how digital financial behavior varies across different consumer segments.

This study aims to bridge the existing research gap by investigating the influence of e-payment usage on consumer behavior and cash management, with a specific focus on the moderating role of financial literacy. The key research objectives are to examine the impact of e-payment usage on consumer spending behavior, particularly in terms of impulse purchases and financial awareness, to analyze the effect of e-payment on cash management by assessing whether digital payment systems enhance financial discipline or lead to weaker financial control, and to evaluate the moderating role of financial literacy in determining whether financially literate individuals mitigate the risks of overspending and maximize the benefits of e-payment for financial planning.

This study makes significant contributions to both theoretical and practical domains. Theoretically, it extends the literature on behavioral economics and financial literacy theory by providing empirical evidence on how e-payment influences financial behavior. Unlike previous studies, this research introduces financial literacy as a key moderating factor, offering new insights into how knowledge and awareness shape digital financial decisions. From a practical perspective, the findings provide valuable implications for policymakers, financial institutions, and e-payment providers. Governments and financial regulators can use these insights to develop targeted financial literacy programs aimed at promoting responsible digital spending habits. Banks and FinTech companies can leverage these findings to design smarter financial management tools, such as spending alerts, automated budgeting features, and personalized financial education within e-wallet applications. Furthermore, consumers can benefit from a better understanding of how e-payment behaviors impact their financial well-being, enabling them to make informed financial choices in an increasingly digital economy.

The remainder of this paper is structured as follows. Section 2 presents the literature review and theoretical framework, outlining key concepts related to e-payment adoption, financial literacy, and

digital financial behavior. Section 3 describes the research methodology, detailing the data collection process, measurement variables, and analytical techniques used in this study. Section 4 provides the results and statistical findings, followed by Section 5, which discusses the implications of the findings in both theoretical and practical contexts. Finally, Section 6 concludes the study, highlighting limitations and future research directions.

LITERATURE REVIEW

A Boundary-Condition View

Previous studies on digital payments have predominantly adopted a linear model, assuming a straightforward relationship between e-payment usage and financial outcomes such as consumer spending or cash management (Usman et al., 2025). While these studies have established that e-payment increases convenience and transaction efficiency, the results regarding its behavioral impact remain inconclusive. Some researchers found that e-payment stimulates impulsive spending by lowering the psychological “pain of paying” (Thaler & Sunstein, 2008), whereas others emphasized its potential to enhance budgeting and cash monitoring through digital transaction records (Liu & Zhang, 2021; Ly & Ly, 2024). These inconsistencies indicate that the linear approach cannot fully explain why e-payment may lead to both overspending and improved financial discipline across different user groups.

This limitation presents a significant research opportunity. Rather than assuming uniform behavioral effects, it is essential to consider conditional factors that may alter how e-payment influences consumer decisions. Prior research rarely integrates psychological and cognitive determinants such as financial knowledge, awareness, and self control into the model (Fernandes et al., 2014; Liu & Zhang, 2021). Consequently, there remains a gap in understanding when and for whom e-payment promotes financial well-being or financial risk.

To address this gap, the present study introduces financial literacy as a moderating construct, positioning it as a boundary condition that determines the strength and direction of e-payment’s behavioral effects. Financial literacy is not only a knowledge-based capability but also a self-regulatory mechanism that enables individuals to interpret financial information, evaluate risks, and make informed digital transactions (OECD, 2018). Individuals with high financial literacy are expected to utilize e-payment tools (e.g., spending dashboards, alerts, and budgeting features) to support better financial control, while those with low literacy may exhibit impulsive or excessive spending patterns.

Theoretically, this research integrates two complementary perspectives. Behavioral Economics Theory explains how reduced transaction friction and lower payment salience in digital systems can distort spending judgment and increase purchase frequency (Thaler & Sunstein, 2008). Meanwhile, Financial Literacy Theory posits that financial knowledge enhances rational decision-making, budgeting, and savings behavior (Lusardi & Mitchell, 2014). By combining these frameworks, the study proposes a moderated theoretical model, where financial literacy serves as a boundary condition that shapes whether e-payment adoption leads to impulsive spending or disciplined cash management.

This integrated model extends previous linear frameworks by explicitly mapping the research gap between technology-driven convenience and human cognitive control, thereby contributing a more nuanced explanation of digital financial behavior in emerging markets such as Indonesia. It also establishes the conceptual foundation for the study’s hypotheses and structural model, which empirically test how financial literacy moderates the dual behavioral effects of e-payment usage.

E-Payment and Consumer Behavior

The adoption of electronic payment (e-payment) systems has transformed how consumers engage in financial transactions, shifting purchasing behaviors from cash-based spending to digital transactions (Dahlberg et al., 2015). Previous studies suggest that e-payment increases transaction frequency and impulse buying by reducing the psychological friction associated with spending money (Hampson et al., 2021). Unlike cash transactions, which involve a tangible exchange of money, digital payments create a psychological detachment from financial loss, making it easier for consumers to make frequent and unplanned purchases. This phenomenon aligns with Behavioral Economics Theory, which argues that

the "pain of paying" is minimized in digital transactions, leading to higher spending tendencies (Thaler & Sunstein, 2008).

Usman et al. (2025) found that the convenience of e-payment systems increases consumer confidence in spending, particularly in the context of online retail and mobile commerce. Similarly, Laukkanen (2016) highlighted that digital payment adoption is driven by ease of use, transaction speed, and perceived security, which encourage consumers to transition from traditional payment methods. However, while e-payment facilitates economic transactions, its influence on spending control and financial awareness remains a subject of debate. Some studies suggest that e-payment can lead to financial mismanagement, particularly among individuals with low financial literacy, as they may engage in excessive discretionary spending without tracking their transactions effectively. A consistent body of work finds that card-based, mobile, and e-wallet payments lower psychological barriers at checkout (reduced salience, deferred pain, one-tap convenience), which is associated with higher basket sizes and greater incidence of impulse purchases compared with cash. This effect is often stronger in settings where payment is instantaneous and authentication is streamlined. Nevertheless, heterogeneity emerges across user segments and contexts, suggesting that individual capabilities and self-regulatory resources can moderate the magnitude of the spending increase. Building on this stream, we treat e-payment usage as a driver of greater spending propensity via friction-reduction channels.

H1. E-payment usage positively affects consumer behavior

E-Payment and Cash Management

In contrast to concerns that digital transactions increase spending, research has also highlighted the positive impact of e-payment on financial management. Liu & Zhang (2021) found that e-payment adoption enables users to track their spending patterns more effectively, as digital transactions provide automated records, categorized expense tracking, and financial alerts. These features enhance budgeting, cash flow monitoring, and long-term financial planning. Unlike cash-based transactions, which require manual tracking, e-payment systems integrate real-time financial data, allowing users to analyze their spending habits and adjust their financial decisions accordingly.

Furthermore, digital transactions encourage structured financial behavior by offering tools such as spending limits, automated bill payments, and financial goal-setting features. Research by Ly & Ly (2024) emphasized that e-payment users exhibit higher financial self-awareness, particularly when leveraging digital banking applications with personalized financial insights. However, the extent to which e-payment enhances financial control depends on user engagement with financial management tools, as individuals who do not actively monitor their transactions may still face financial discipline challenges. Beyond convenience, e-payment ecosystems supply automated tracking (timestamps, merchant/category tags, dashboards, alerts) that can scaffold budgeting, reconciliation, and bill planning. Empirical studies link digital transaction records to improved monitoring and planning behaviors, although the strength of these benefits depends on how effectively users interpret and act on the information surfaced by apps. We therefore conceptualize e-payment usage as enabling better cash management through information visibility and routine prompts.

H2. E-payment usage positively affects cash management

The Moderating Role of Financial Literacy

Financial literacy has been widely recognized as a critical factor in financial decision-making, influencing how individuals manage their expenses, savings, and investment choices. According to Financial Literacy Theory Lusardi & Mitchell (2014), individuals with higher financial knowledge are more likely to engage in responsible financial behaviors, such as budgeting, expense tracking, and informed spending decisions. In the context of e-payment adoption, financial literacy serves as a moderating factor that determines whether digital transactions lead to financial discipline or excessive spending (OECD, 2018).

Zhavoronok et al. (2022) found that financially literate individuals are more likely to utilize e-payment tools strategically, leveraging digital tracking features to optimize their cash flow and spending patterns.

Conversely, individuals with low financial literacy may experience greater difficulty managing digital transactions, as they may fail to distinguish between essential and non-essential spending. [Laukkanen \(2016\)](#) similarly observed that financial literacy influences how consumers perceive and interact with digital financial services, shaping their ability to navigate complex financial products effectively.

This study builds on prior research by investigating whether financial literacy mitigates the impulsive spending tendencies associated with e-payment while enhancing cash management capabilities. By analyzing financial literacy as a moderating variable, this research provides insights into how consumer financial behavior varies across different levels of financial knowledge, contributing to the broader discussion on responsible digital financial management.

H3. Financial literacy attenuates the positive effect of e-payment usage on spending

H4. Financial literacy attenuates the positive effect of e-payment usage on cash management

METHODS

Research Design

This study employs a quantitative research approach to examine the causal relationships between e-payment usage, consumer behavior, and cash management, with financial literacy as a moderating variable. A survey-based research design was utilized to systematically collect and analyze empirical data, ensuring objectivity in measuring the proposed relationships. A structured questionnaire was distributed to residents of Central Java who actively use electronic payment systems. The study follows a positivist paradigm, enabling statistical generalization of the findings to a broader population. Through hypothesis testing, this research aims to provide empirical evidence on how e-payment adoption influences financial behavior and how financial literacy moderates these effects ([Lusardi & Mitchell, 2014](#); [Putri et al., 2024](#)).

Research Location

The study was conducted in Central Java, Indonesia, a region experiencing rapid growth in digital financial services. The increasing government initiatives promoting cashless transactions make Central Java a relevant case study for understanding the behavioral and financial implications of e-payment adoption ([Wiranatakusuma & Latief, 2024](#)). The study focused on individuals who regularly engage in digital transactions, as they provide valuable insights into the influence of electronic payments on consumption patterns and personal cash management.

Population and Sampling

The target population comprises individuals in Central Java who use e-payment systems for daily transactions. Given the study's objectives, a purposive sampling technique was employed to ensure that participants met specific criteria. Eligible respondents were required to be at least 18 years old, ensuring that they had financial decision-making autonomy ([Lusardi & Mitchell, 2014](#)), and to be active users of e-payment services, including mobile banking, e-wallets, or digital payment platforms.

To determine the appropriate sample size, Slovin's formula was applied as follows:

$$\eta = \frac{N}{1 + Ne^2}$$

where η represents the required sample size, N is the total population, and e is the margin of error. With a 95% confidence level ($\alpha=0.05$), the estimated sample size for this study was 400 respondents. The selection of this sample size ensures sufficient statistical power to detect meaningful relationships between variables ([Sugiyono, 2017](#)).

Measurement of Variables

Data were collected using a structured questionnaire designed to assess key research variables. Respondents provided their answers using a five-point Likert scale ranging from 1 (Strongly Disagree)

to 5 (Strongly Agree). The questionnaire was adapted from previously validated studies to ensure construct validity and reliability.

The study measured four primary variables: e-payment usage, consumer behavior, cash management, and financial literacy. E-payment usage was assessed based on transaction frequency, convenience, security, and promotional benefits. Consumer behavior focused on shopping frequency, impulsivity, spending awareness, and online shopping accessibility. Cash management was evaluated through expense tracking, transaction monitoring, budgeting, and spending control. Financial literacy measured respondents' understanding of e-payment risks, financial planning habits, and budgeting discipline.

Table 1. Research Statement Instrument

Variable	Statement	Source
E-Payment (X)	I often use e-payment in my daily transactions. I find e-payment easier to use compared to cash. I prefer e-payment because transactions are faster. I feel safe using e-payment for shopping. I often use the promotion or cashback features in e-payments.	(Ly & Ly, 2024)
Consumer Behavior (Y1)	I tend to shop more often since using e-payment. I am more impulsive in purchasing goods when using e-payment. I feel like i dont really pay attention to the amount of spending when using e-payment. I find it easier to access online shopping services with e-payment. I often use e-payment for non-essential transactions (entertainment, fast food, etc.)	(Hampson et al., 2021)
Cash Management (Y2)	I always record my expenses when using e-payment. I find it easier to manage my finances with the transaction history feature in e-payment. I set a monthly spending limit for using e-payment. I often evaluate my expenses based on e-payment transaction data. I feel that using e-payment helps me control my spending.	(Liu & Zhang, 2021)
Financial literacy (Z)	I understand how e-payment works and the risk. I know how to manage personal finances well. I always consider the financial impact before making an e-payment transaction. I have a habit of making a budget before shopping. I am aware of the importance of saving even though I often use e-payments.	(Laukkanen, 2016)

Source: Processed Data (2025)

Pre-Test and Instrument Validation

Before conducting the full-scale survey, a pre-test was carried out with 67 respondents to evaluate the clarity, reliability, and validity of the questionnaire. Based on pre-test feedback, minor revisions were made to ensure that all items were clearly understood by respondents (Presser et al., 2004). Additionally, a Focus Group Discussion (FGD) was conducted with financial experts and researchers to refine ambiguous items, ensuring that the instrument accurately captured respondents' perceptions (Krueger, 2014).

To ensure the validity and reliability of the measurement instruments, multiple statistical tests were performed. Convergent validity was assessed using Average Variance Extracted (AVE), with a minimum threshold of 0.50 to confirm that the items within each construct were sufficiently correlated. Discriminant validity was examined using the Fornell-Larcker criterion and the Heterotrait-Monotrait (HTMT) ratio, ensuring that constructs were empirically distinct from one another (Henseler et al., 2015). Reliability analysis was conducted using Cronbach's Alpha and Composite Reliability (CR), with a threshold of 0.70 or higher, indicating consistent internal reliability across the measured variables.

Data Analysis Technique

The study employed Partial Least Squares - Structural Equation Modeling (PLS-SEM) for data analysis, using SmartPLS 4 software. This method was chosen due to its capability to handle complex models with latent constructs and its suitability for analyzing moderating effects (Hair et al., 2021; Sarstedt et al., 2014).

Data analysis began with an examination of the measurement model (outer model) to verify that the observed indicators accurately reflected their respective latent constructs. Factor loadings were assessed to ensure that each indicator contributed significantly to its construct (Hair et al., 2021). The structural model (inner model) was then evaluated to examine the strength and direction of the relationships between variables. Path coefficients were analyzed to determine the statistical significance of the proposed hypotheses, while the coefficient of determination (R^2) was used to measure the proportion of variance explained by the independent variables. Effect size (f^2) was also calculated to assess the practical significance of predictor variables.

Hypothesis testing was conducted using Bootstrapping with 5,000 resamples to determine the significance of relationships within the model. Hypotheses were considered statistically supported if the t-statistic exceeded 1.96 and the p-value was less than 0.05 (Henseler et al., 2015). Furthermore, a moderation analysis was performed to examine whether financial literacy strengthened or weakened the relationships between e-payment usage, consumer behavior, and cash management (Chin, 2009; Sarstedt et al., 2019).

FINDINGS

Respondent Profile

The demographic characteristics of the respondents are summarized in Table 2. The majority of respondents were female (62.4%), suggesting that women play a significant role in managing personal finances and engaging with e-payment systems. Most respondents were aged 25–34 years (48.7%), followed by those aged 35–44 years (32.1%), indicating that the productive age group dominates e-payment usage. Regarding education, 45.2% of respondents held a bachelor's degree, while 28.4% completed senior high school, reflecting a relatively high level of financial literacy.

Table 2. Demographic Characteristics of Respondents

Characteristic	Category	Frequency	Percentage (%)
Gender	Male	151	37.6
	Female	249	62.4
Age	18–24 years	72	18.0
	25–34 years	195	48.7
	35–44 years	129	32.1
	45+ years	14	3.5
	High School	114	28.4
Education Level	Diploma (D3)	67	16.7
	Bachelor's Degree (S1)	181	45.2
	Master's Degree (S2)	38	9.5
Monthly Income (IDR)	< 3 million	96	24.0
	3–6 million	214	53.5
	> 6 million	90	22.5

Sources: Processed Data (2025)

Descriptive Statistics

Table 3 reports the descriptive statistics for the study constructs. Respondents exhibit moderately high e-payment usage ($M = 3.68$, $SD = 0.94$) and spending propensity ($M = 3.97$, $SD = 0.97$), alongside comparable cash-management practices ($M = 3.83$, $SD = 0.93$). Financial literacy shows a moderate central tendency ($M = 3.46$, $SD = 1.04$). Minimum–maximum values span the full Likert continuum, indicating adequate response variability. Skewness and kurtosis remain within acceptable ranges, suggesting no severe departures from normality for large-sample SEM.

Table 3. Descriptive Statistics of Study Variables

Variable	k	Scale	Mean	SD	Min	Max
E-Payment (X)	5	1-5	3.68	0.94	1.20	5.00
Financial Literacy (Z)	5	1-5	3.46	1.04	1.40	5.00
Consumer Behavior (Y1)	5	1-5	3.97	0.97	1.80	5.00
Cash Management (Y2)	5	1-5	3.83	0.93	1.60	5.00

Source: Processed Data (2025)

Measurement Model Analysis

The measurement model was assessed using factor loadings, reliability, and validity tests. Table 4 presents the reliability and validity results, confirming that all constructs met the required thresholds. Cronbach's Alpha values exceeded 0.70, indicating strong internal consistency. Average Variance Extracted (AVE) values were above 0.50, demonstrating sufficient convergent validity.

Table 4. Reliability and Validity of Measurement Model

	Cronbach's alpha	Composite reliability (rho_a)	Composite reliability (rho_c)	Average variance extracted (AVE)
E Payment (X)	0.885	0.897	0.916	0.686
Financial Literacy (Z)	0.872	0.913	0.905	0.656
Consumer Behavior (Y1)	0.905	0.923	0.929	0.722
Cash Management (Y2)	0.885	0.896	0.916	0.685

Source: Processed Data (2025)

Structural Model Analysis

Inner Model

Based on the results of the inner model analysis using the sem-pls approach, the r-square value for consumer behavior (y1) is 0.877, while for cash management (y2) it is 0.831 (see Table 5). These values indicate that the model has a very strong explanatory power, as both exceed the 0.75 threshold commonly referenced in the literature (Chin, 2009). Specifically, 87.7% of the variance in consumer behavior can be explained by the independent variables included in the model, while 83.1% of the variance in cash management is similarly accounted for. This reflects a high degree of model fit and suggests that the structural model is capable of effectively capturing the relationships among the latent constructs. Furthermore, the adjusted r-square values (0.876 for y1 and 0.830 for y2) are nearly identical to the unadjusted values, indicating that the model is stable and not overfitted, despite the inclusion of multiple predictors. Overall, these findings demonstrate that the model has strong predictive relevance and is robust for explaining the hypothesized relationships.

Table 5. R-Square and Goodness Fit

	R-square	R-square adjusted
Cash Management (Y2)	0.831	0.830
Consumer Behavior (Y1)	0.877	0.876
	Saturated model	Estimated model
SRMR	0.035	0.036
d_ULS	0.252	0.275
d_G	0.961	0.978
Chi-square	1916.207	1923.370
NFI	0.849	0.849

Sources: Processed Data (2025)

The Influence of E-Payment on Consumer Behavior

The results of the PLS-SEM analysis indicate that e-payment usage has a significant positive effect on consumer behavior ($p < 0.000$) (see Table 6). This suggests that increased reliance on digital payment methods encourages higher spending, supporting previous research on impulse buying and digital convenience (Thaler & Sunstein, 2008).

The Influence of E-Payment on Cash Management

E-payment usage also shows a significant positive relationship with cash management ($p < 0.000$), implying that electronic transactions improve financial monitoring and tracking. This finding aligns with [Liu & Zhang \(2021\)](#), who highlighted the role of digital payments in financial management efficiency.

The Moderating Role of Financial Literacy

The moderation analysis reveals that financial literacy weakens the positive relationship between e-payment usage and consumer behavior ($p < 0.000$). This indicates that individuals with higher financial literacy tend to exercise greater control over their spending, even when using e-payment systems. Conversely, financial literacy strengthens the relationship between e-payment usage and cash management ($p < 0.005$), suggesting that financially literate individuals utilize e-payment features to enhance their financial planning and monitoring.

Table 6. Structural Model Path Coefficients

	Original sample (O)	Sample mean (M)	Standard deviation (STDEV)	T statistics (O/STDEV)	P values
E-Payment (X) -> Consumer Behavior (Y1)	0.731	0.729	0.045	16.153	0.000
E-Payment (X) -> Cash Management (Y2)	0.493	0.494	0.051	9.597	0.000
Financial Literacy (Z) x E-Payment (X) -> Consumer Behavior (Y1)	-0.082	-0.083	0.021	3.880	0.000
Financial Literacy (Z) x E-Payment (X) -> Cash Management (Y2)	-0.065	-0.065	0.023	2.826	0.005

Sources: Processed Data (2025)

Graphical Representation of Results

Figure 2 presents the structural model results, highlighting the significant relationships among e-payment usage, consumer behavior, cash management, and financial literacy moderation effects. The path coefficients and significance levels illustrate the influence of financial literacy in shaping financial behaviors.

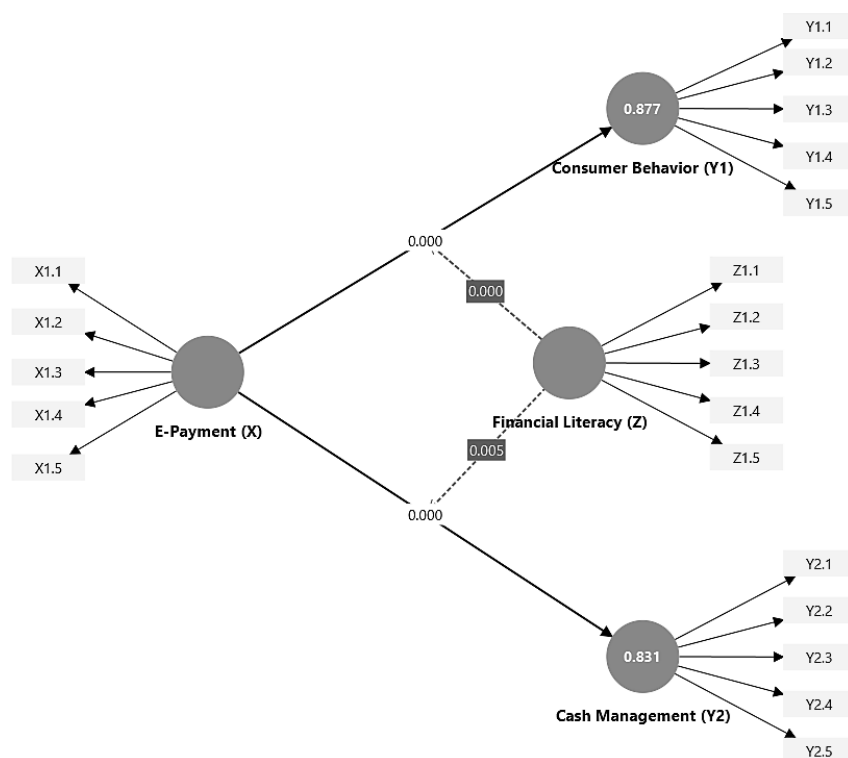


Figure 2. Structural Model Results (PLS-SEM Analysis)

Sources: Processed Data (2025)

DISCUSSION

To the best of our knowledge, this is the first study to comprehensively examine the dual role of e-payment usage in shaping consumer behavior and cash management, while also investigating financial literacy as a moderating factor. By employing PLS-SEM analysis on a large sample of respondents in Central Java, Indonesia, this study provides empirical evidence that, while e-payment adoption increases spending behavior, it also enhances financial management efficiency. Furthermore, our findings highlight that financial literacy serves as a critical buffer, mitigating the risk of excessive spending and maximizing the benefits of digital financial tools. These results contribute to the ongoing discourse in behavioral economics, digital finance, and financial literacy research, offering both theoretical advancements and practical recommendations for consumers, policymakers, and financial institutions.

Interpretation of Key Findings

The Influence of E-Payment on Consumer Behavior

The findings reveal that e-payment usage significantly increases consumer spending, supporting the notion that digital payment systems encourage higher transaction frequency and impulse purchasing. This result aligns with Behavioral Economics Theory (Thaler & Sunstein, 2008), which posits that individuals experience less "pain of paying" when using digital payment methods compared to cash. The psychological detachment from physical money reduces the cognitive awareness of expenditure, leading consumers to spend more effortlessly (Prelec & Loewenstein, 1998).

These findings are consistent with prior research on digital payment adoption and impulse buying behavior (Usman et al., 2025). The frictionless nature of e-payment, combined with instant transaction processing, alters financial decision-making patterns, making consumers more likely to make unplanned purchases. Moreover, e-payment platforms often incorporate reward-based mechanisms, such as cashback incentives and promotional discounts, which further reinforce spending behavior (Liébana-Cabanillas et al., 2018). The instant gratification effect associated with digital transactions creates a habit loop, where frequent users become conditioned to making purchases with minimal financial reflection.

However, the impact of e-payment on spending behavior is not uniform across all users. The results indicate that financially literate individuals exhibit greater spending control despite using e-payment systems. This suggests that the ability to critically evaluate financial decisions serves as a protective mechanism, preventing users from engaging in excessive or impulsive spending (Lusardi & Mitchell, 2014). Unlike financially illiterate individuals, who may perceive e-payment as a tool for convenience and consumption, financially literate users are more likely to view digital transactions as part of a broader financial strategy.

The Influence of E-Payment on Cash Management

In contrast to concerns that e-payment reduces financial discipline, our findings suggest that e-payment usage positively affects cash management, enhancing an individual's ability to track, analyze, and regulate financial transactions. This result aligns with Liu & Zhang (2021), who argued that digital financial tools improve expenditure monitoring and budgeting accuracy. Unlike cash transactions, which lack automated tracking features, e-payment systems provide real-time expenditure records, categorized spending breakdowns, and predictive financial analytics, enabling users to develop more structured financial habits (Liébana-Cabanillas et al., 2018).

Our findings indicate that e-payment users engage in more systematic financial planning, leveraging digital tracking tools to enhance financial awareness. The availability of transaction histories allows users to identify spending patterns, set budgeting constraints, and adjust financial decisions accordingly (Mora, 2022). This contradicts the traditional belief that cash-based spending leads to superior financial management; rather, digital transaction records facilitate a data-driven approach to personal finance, allowing individuals to make informed decisions based on actual expenditure trends (Lusardi & Mitchell, 2014).

However, potential drawbacks exist. Over-reliance on digital financial tracking may reduce manual financial discipline, as users may become less engaged in active financial decision-making and instead rely solely on automated tools for budgeting. This aligns with the concern raised by [Alshebami & Al Marri \(2022\)](#) that digital financial platforms may create a false sense of financial control, leading users to overlook long-term financial planning considerations.

The Moderating Role of Financial Literacy

A key contribution of this study is the identification of financial literacy as a moderating factor that influences the relationship between e-payment, consumer behavior, and cash management. The results indicate that higher financial literacy weakens the positive relationship between e-payment and consumer spending, meaning that individuals with greater financial knowledge are less susceptible to the impulsive spending tendencies associated with digital transactions. This finding aligns with Financial Literacy Theory ([Lusardi & Mitchell, 2014](#)), which posits that financial knowledge improves self-control, decision-making, and long-term financial planning.

Our findings corroborate prior studies that suggest financially literate individuals are more likely to use budgeting tools effectively ([Fernandes et al., 2014](#); [Zhavoronok et al., 2022](#)). Unlike financially uneducated users, who may use e-payment passively, financially literate users actively engage with financial management features, such as spending categorization, automated budgeting, and transaction alerts. This enhances their ability to regulate expenses, optimize savings, and make informed financial choices.

Furthermore, financial literacy strengthens the positive effect of e-payment on cash management, suggesting that knowledgeable individuals maximize the financial advantages offered by digital financial systems. While e-payment itself provides tools for financial management, its effectiveness depends on the user's ability to interpret and utilize financial information strategically ([OECD, 2018](#)).

CONCLUSION

This study provides empirical evidence on the dual influence of e-payment usage on consumer behavior and cash management, highlighting the moderating role of financial literacy. The findings indicate that digital payment systems, while reducing the psychological barriers to spending and encouraging impulsive purchases, also enhance users' ability to monitor, plan, and manage their finances through automated tracking and budgeting features. Crucially, financial literacy shapes these behavioral outcomes: individuals with higher literacy demonstrate stronger self-control and use digital tools more effectively for financial discipline, whereas those with lower literacy are more prone to impulsive and unplanned spending.

Theoretically, this study enriches the literature on behavioral finance, consumer psychology, and financial literacy by introducing a moderated model that integrates behavioral, cognitive, and technological perspectives. The results extend Behavioral Economics Theory by confirming that the reduced "pain of paying" in digital transactions drives higher spending tendencies. At the same time, they strengthen Financial Literacy Theory by positioning literacy not merely as a form of knowledge but as a self-regulatory mechanism that mitigates financial risks and enhances decision-making in digital environments. This study thus challenges the conventional assumption that e-payment inherently undermines financial discipline, showing instead that its impact depends on the interplay between financial literacy and user self-regulation.

Practically, the findings offer valuable insights for multiple stakeholders. For consumers, cultivating digital financial literacy is essential to optimize the benefits of e-payment while maintaining spending control. For policymakers, the results underscore the importance of integrating financial education and digital literacy initiatives within broader financial inclusion programs. Governments and financial institutions should collaborate to design targeted interventions that foster responsible digital financial behavior. For e-payment providers and banks, the study highlights opportunities to embed behavioral design features such as spending alerts, automated budgeting recommendations, and gamified savings incentives into digital platforms to promote user engagement and financial well-being.

Despite its contributions, the study is subject to several limitations. Its geographical focus on Central Java restricts the generalizability of findings to regions with different financial cultures and technological infrastructures. The reliance on self-reported data may also introduce social desirability bias, leading to underreporting of impulsive behavior. Furthermore, the study concentrates primarily on financial literacy as a moderating factor, excluding other behavioral and contextual variables such as income, digital literacy, and psychological biases. Future research should employ cross-regional and longitudinal designs to capture behavioral dynamics over time and examine emerging influences such as AI-based financial advisors, cashback mechanisms, and gamified saving systems.

Overall, this study advances theoretical and practical understanding of digital financial behavior by demonstrating that e-payment is neither inherently beneficial nor detrimental. Its effects are contingent upon users' financial literacy and capacity for self-regulation. Strengthening digital financial literacy, therefore, represents a crucial step toward ensuring that technological innovation contributes to more responsible and sustainable financial practices in the digital economy.

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CONFLICT OF INTEREST STATEMENT

The author declares that there are no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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